

LM06 Analysis of Inventories

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1. Introduction

Inventories are assets held by a company to produce finished goods for sale. They are shown as a current asset on the balance sheet; and can represent a significant part of the total assets for many companies.

Manufacturing and merchandising companies (Ex: Nike, Caterpillar) generate sales and profit through the sale of inventory. An important measure in calculating profits is cost of goods sold, i.e., how much cost the company incurred from procuring raw materials to converting it to a finished product, and finally selling it.

There is no universal inventory valuation method. The four inventory valuation methods used by companies are:

First In First Out (FIFO)

- The cost of the first item purchased is assumed to be the cost of the first item sold.
- Ending inventory is based on the cost of the most recent purchases.

Last In First Out (LIFO)

- The cost of the last item purchased is assumed to be the cost of the first item sold.
- Ending inventory is based on the cost of the earliest purchases.

Weighted average cost

- Each item in the inventory is valued using an average cost of all items in the inventory.
- COGS and inventory values are between their FIFO and LIFO values.

Specific identification

- Each unit sold is matched with the unit's actual cost.
- This method is usually used for items that are unique in nature, for example, jewelry.

All four methods are permitted under U.S. GAAP. However, IFRS does not permit LIFO method.

2. Inventory Valuation

Holding inventory for a prolonged period results in the risk of spoilage, obsolescence, or decline in prices, and the cost of inventory may not be recoverable in such circumstances. We define some terms first before looking at the differences in how inventory is measured under IFRS and GAAP.

Net realizable value: Estimated selling price under ordinary business conditions minus estimated costs necessary to get the inventory in condition for sale. NRV is from a seller's perspective.

Net realizable value = estimated sales price – estimated selling costs

Market value: Current replacement cost subject to lower or upper limits. Market value has

upper limit of net realizable value and lower limit of NRV less a normal profit margin. Market value is from a buyer's perspective.

Market value limits = (NRV - normal profit margin, NRV)

The following table describes how inventory is measured under IFRS and GAAP:

<i>Inventory measurement under IFRS and US GAAP</i>	
<i>IFRS</i>	<i>US GAAP</i>
Lower of cost or net realizable value.	Lower of cost or market value.
If NRV is less than the balance sheet cost, the inventory is "written down" to NRV. The loss in value is reflected in the income statement in cost of goods sold. Inventory write-down has a negative effect on profitability, liquidity, and solvency ratios and positive effect on activity ratios.	If cost exceeds market, inventory is written down to market value on the balance sheet and the loss is recognized.
If value recovers subsequently, inventory can be written up and gain is recognized in the income statement. The amount of gain is limited to loss previously recognized.	If value recovers subsequently, no write up is allowed. There is no reversal of write-downs. This may motivate companies not to record inventory write-downs unless the decline is permanent as it affects profitability ratios.
Commodities and agricultural goods prices can be reported above historical cost.	Commodities and agricultural goods prices can be reported above historical cost.

An inventory write-down reduces both profit and carrying amount of inventory on the balance sheet, which, in turn, affects the ratios. The following table shows the effect of inventory write-downs on various financial ratios:

<i>Ratio</i>	<i>Effect</i>	<i>Reason</i>
<i>Liquidity ratios</i>		
Current ratio	Lower	Current assets decrease due to lower inventory.
<i>Activity ratios</i>		
Inventory turnover	Higher	COGS increases assuming inventory write-downs are reported as part of cost of sales. Average inventory decreases. Lower inventory carrying amounts make it appear as if the company is managing its inventory effectively, but write-downs reflect poor inventory management.

Days of inventory on hand	Lower	Inventory turnover is higher.
<i>Profitability ratios</i>		
Net profit margin	Lower	Cost of sales is higher. Sales stay the same.
Gross profit margin	Lower	Cost of sales is higher. Sales stay the same.

Companies that use weighted average, specific identification, and FIFO are more likely to have inventory write-downs than companies using the LIFO method.

3. The Effects of Inflation and Deflation on Inventories, Costs of Sales, and Gross Margin

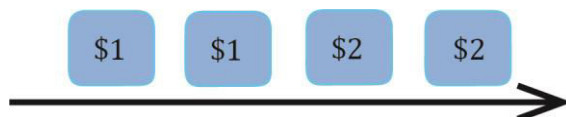
First In, First Out (FIFO)

Under First In, First Out:

- Oldest goods purchased or manufactured are assumed to be sold first.
- Newest goods purchased or manufactured remain in ending inventory.
- When prices are increasing or stable, cost assigned to items in inventory is higher than the cost of items sold.

The following example illustrates how cost of goods sold and inventory are accounted for in each period:

Assume you bought four pencils. The first two pencils were worth \$1 each and the next two pencils were worth \$2 each. Before you start selling, your inventory consists of four pencils.



In period 1, you sell two pencils. The cost of pencils sold in period 1 is \$2 (two pencils of \$1 each). The pencils that were bought first are considered sold. Inventory at the end of period 1 is \$4 and looks like this (2 pencils of \$2 each):



As you could see, cost of pencils sold in period 1 was \$2 (cheaper pencils bought initially) whereas the cost of pencils in inventory was \$4. In period 2, you again sell two pencils. The cost of pencils sold in period 2 is \$4. Inventory at the end of period 2 is 0.

Advantage of using FIFO is that it is less subject to manipulation. The ending inventory is valued based on most recent purchases. COGS is based on earliest purchase costs. Therefore, in an inflationary environment, FIFO results in higher net income.

Weighted Average Cost

Under weighted average cost method, each item in inventory is valued using an average cost

of all items in the inventory.

$$\text{Weighted average cost} = \frac{\text{Total cost of units available for sale}}{\text{Total units available for sale}}$$

Let's use the pencils example again to illustrate how inventory is calculated using the WAC method.

Total cost of pencils available for sales = \$6

Total number of pencils available for sale = 4

Weighted average cost per pencil = \$6/4 = \$1.5

<i>WAC Method</i>	
<i>Item</i>	<i>WAC</i>
Cost of 2 pencils sold in period 1	\$3
Inventory for 2 pencils at the end of period 1	\$3
Cost of 2 pencils sold in period 2	\$3
Inventory at the end of period 2	\$0

Last In, First Out (LIFO)

Under Last In, First Out method:

- The newest items purchased or manufactured are assumed to be sold first.
- Oldest goods purchased or manufactured remain in ending inventory.
- The cost of goods sold reflects the cost of goods purchased or manufactured recently; the value of inventory reflects the cost of older goods purchased.

Let's continue with the pencils example to see how inventory is accounted for in LIFO:

Unlike FIFO, at the end of period 1, LIFO inventory consists of the first two pencils:



<i>LIFO Method</i>	
<i>Item</i>	<i>LIFO</i>
Cost of 2 pencils sold in period 1	\$4
Inventory for 2 pencils at the end of period 1	\$2
Cost of 2 pencils sold in period 2	\$2
Inventory at the end of period 2	\$0

LIFO is not allowed under IFRS; it is allowed only under US GAAP. Companies use LIFO during inflation to reduce taxes as cost of goods sold (COGS) is high.

Under LIFO, ending inventory is valued using earliest purchase costs. Therefore, in an inflationary environment, LIFO ending inventory will be less than current costs. Also, LIFO COGS will be higher than FIFO COGS leading to a lower net income.

Calculation of Cost of Sales, Gross Profit, and Ending Inventory

Based on the inventory valuation method used by a company, the allocation of inventory costs between cost of goods sold on the income statement and inventory on the balance sheet varies in periods of changing prices.

Continuing with the pencils example, assume each of the pencils was sold for \$5. The table below summarizes the cost of goods sold, inventory ending value, and gross profit under each of the methods:

<i>Inventory Accounting under Various Methods</i>			
Item	FIFO (in \$)	LIFO (in \$)	WAC (in \$)
COGS for period 1	2	4	3
Gross profit for period 1	8	6	7
Inventory at end of period 1	4	2	3
COGS for period 2	4	2	3
Gross profit for period 2	6	8	7
Inventory at end of period 2	0	0	0

Some points to be noted:

- The total gross profit and COGS for all the periods combined is the same under each of the methods.
- As the prices of pencils were increasing, the ending inventory was highest and COGS was lowest under FIFO.
- Similarly, the ending inventory was lowest and COGS was highest under LIFO.

The Effects of Inflation and Deflation

The allocation of total cost of goods available for sale to COGS and ending inventory varies under different inventory valuation methods. The following table compares LIFO vs. FIFO for different parameters when prices are rising and inventory levels are stable:

<i>LIFO vs. FIFO with rising prices and stable inventory levels</i>		
	<i>LIFO</i>	<i>FIFO</i>
COGS	Higher	Lower
Taxes	Lower	Higher
Earnings before taxes (EBT)	Lower	Higher
Earnings after taxes (Net Income)	Lower	Higher
Ending inventory	Lower	Higher

Working capital (CA-CL)	Lower	Higher
Cash flow (after tax)	Higher	Lower

Instructor's Note:

Weighted average costs provide results between FIFO and LIFO. Some tips for remembering the table above are listed below:

1. Remember the pencils example of \$1, \$1, \$2, and \$2. Deducing the LIFO values from this example for COGS, net income, and ending inventory becomes simpler.
2. FIFO is the opposite of LIFO.
3. Cash flow (after tax) is higher under LIFO as taxes paid are lower.
4. Companies following US GAAP prefer LIFO because the taxes paid are lower.
5. LIFO gives a better income statement and FIFO a better balance sheet as they reflect economic reality or recent costs. Under LIFO, cost of goods sold in income statement shows the most recent costs reflecting better quality. Similarly, under FIFO ending inventory on the balance sheet shows the most recent costs, reflecting better quality.

LIFO Reserve

The LIFO reserve is the difference between the reported LIFO inventory carrying amount and the inventory amount that would have been reported if the FIFO method had been used instead. The equation for LIFO reserve is given by:

$$\text{LIFO reserve} = \text{FIFO inventory value} - \text{LIFO inventory value}$$

LIFO Liquidation

In periods of rising inventory, the carrying amount of inventory under FIFO will exceed the carrying amount of inventory under LIFO. LIFO reserve is equal to the difference between LIFO inventory and FIFO inventory. LIFO reserve may increase for two reasons:

- The number of inventory units manufactured or purchased exceeds the number of units sold.
- Increasing difference between the older costs used to value inventory under LIFO and the more recent costs used to value inventory under FIFO.

If a firm is liquidating its inventory or if the prices are declining, the LIFO reserve will decline.

When the number of units sold in a period exceeds the number of units purchased/manufactured, it is called 'LIFO liquidation'. In LIFO liquidation, the costs from older LIFO layers will flow to COGS and it can be used by the management to manipulate earnings and margins. The gross profits increase because the older inventory carrying amounts are used for COGS while sales are at current prices. An increase in gross profit accompanied by a decrease in LIFO reserve must be used as a warning sign.

4. Presentation and Disclosure

The efficiency and effectiveness of inventory management can be evaluated using the following ratios:

- Inventory Turnover
- Days of Inventory on hand
- Gross Profit Margin

An analyst must understand that the choice of inventory valuation method can impact several financial ratios and make comparisons between two firms difficult. He needs to be particularly careful when comparing an IFRS and US GAAP firm.

Presentation and Disclosure

IFRS requires the following financial statement disclosures concerning inventory:

- The accounting policies used to measure inventory, including the cost formula.
- The total carrying amount of inventories and the carrying amount in classification (for example, merchandise, raw materials, production supplies, work in progress, and finished goods appropriate to entity).
- The carrying amount of inventories carried at fair value less costs to sell.
- The amount of inventories recognized as an expense in the period (cost of sales).
- The amount of any reversal of any write-down recognized as a reduction in cost of sales in the period.
- What led to the reversal of a write-down in the inventories?
- Carrying amount of inventories pledged as a security for liabilities.

Disclosures under U.S. GAAP are similar to IFRS except that it does not permit reversal of write down of inventories. In addition, any income from liquidation of LIFO inventory must be disclosed.

Inventory Ratios

The choice of inventory valuation method impacts various components of the financial statements such as cost of goods sold, net income, current assets, and total assets. As a result, it affects the financial ratios containing these items. Analysts must consider the differences in valuation methods when evaluating a company's performance over time or in comparison to other companies.

The table below summarizes the impact of valuation method on inventory-related ratios in an inflationary environment:

<i>Ratio</i>	<i>Numerator</i>	<i>Denominator</i>	<i>Impact on ratio</i>
Inventory turnover	Cost of goods sold is higher under LIFO.	Average inventory is lower under LIFO.	Higher under LIFO.
Days of inventory	No. of days are the same.	Higher under LIFO.	Lower under LIFO.
Total asset turnover	Revenue is the same.	Lower average total assets under LIFO.	Higher under LIFO.
Current ratio	Ending inventory is lower under LIFO so current assets are lower.	Current liabilities are the same.	Lower under LIFO.
Cash ratio	Cash is higher under LIFO because taxes are lower.	Current liabilities are the same.	Higher under LIFO.
Gross profit margin	Gross profit is lower under LIFO as COGS is higher.	Revenue is the same.	Lower under LIFO.
Return on assets	Net income is lower under LIFO as COGS is higher.	Lower average total assets under LIFO.	Lower under LIFO.
Debt to equity	Debt is the same.	Lower equity under LIFO. $\text{Equity} = \text{assets} - \text{liabilities}$. Total assets under LIFO are lower as ending inventory is lower.	Higher under LIFO.

The ratios that are important in evaluating a company's management of inventory are inventory turnover, number of days of inventory, and gross profit margin. A high inventory turnover implies that a company is utilizing inventory efficiently.

Inventory Analysis

Retailers normally report inventory in a single account. Whereas, manufacturing companies usually classify inventory into three separate accounts: raw materials, work in progress, and finished goods. These classifications can provide insights into a company's future sales and profits.

For example, a significant increase in the raw materials or work in progress inventory may be considered as a sign of increased demand, and higher future sales and profits. On the other hand, a significant increase in the finished goods inventory may be considered as a sign of slowing demand, and lower future sales and profits.

Analysts should also compare the growth rate of finished goods inventory to the growth rate

of sales. For example, if growth of inventories is greater than the growth of sales, this could indicate slowing demand.

Summary

LO: Describe the measurement of inventory at the lower of cost and net realizable value and its implications for financial statements and ratios.

Net realizable value: It is calculated as estimated selling price under ordinary business conditions minus estimated costs necessary to get the inventory in condition for sale.

Net realizable value = estimated sales price – estimated selling costs

Market value: It is the current replacement cost subject to lower or upper limits. Market value has upper limit of net realizable value and lower limit of NRV less a normal profit margin.

Market value limits = (NRV - normal profit margin, NRV)

Under IFRS, inventories are valued at the lower of cost or net realizable value. Inventory write-ups are allowed but only to the extent a previous write-down to net realizable value was recorded.

Under US GAAP, inventories are valued at lower of cost or market. If cost exceeds market, inventory is written down. No subsequent write-ups are allowed.

When inventory is written down from cost to net realizable value:

- It decreases inventory, assets, and equity.
- It increases asset turnover, debt to equity ratio, and the debt to assets ratio.
- It results in a loss on the income statement, which decreases net income, net profit margin, return on assets, and return on equity.

LO: Calculate and explain how inflation and deflation of inventory costs affect the financial statements and ratios of companies that use different inventory valuation methods.

The following table compares LIFO and FIFO when prices are rising and inventory levels are stable. (We get the opposite effect during periods of falling prices)

<i>LIFO vs. FIFO with rising prices and stable inventory levels</i>		
	<i>LIFO</i>	<i>FIFO</i>
COGS	Higher	Lower
Taxes	Lower	Higher
Earnings before taxes (EBT)	Lower	Higher
Earnings after taxes (Net Income)	Lower	Higher
Ending inventory	Lower	Higher
Working capital (CA-CL)	Lower	Higher
Cash flow (after tax)	Higher	Lower

For weighted average costs, all values will be between those for the LIFO and FIFO methods.

LO: Describe the presentation and disclosures relating to inventories and explain issues that analysts should consider when examining a company's inventory disclosures and other sources of information.

IFRS requires the following financial statement disclosures concerning inventory:

- The accounting policies used to measure inventory, including the cost formula.
- The total carrying amount of inventories and the carrying amount in classification.
- The carrying amount of inventories carried at fair value less costs to sell.
- The amount of inventories recognized as an expense in the period (cost of sales).
- The amount of any reversal of any write-down recognized as a reduction in cost of sales in the period.
- What led to the reversal of a write-down in the inventories?
- Carrying amount of inventories pledged as a security for liabilities.

Disclosures under U.S. GAAP are similar to IFRS except that U.S. GAAP does not permit reversal of write-down of inventories. In addition, any income from liquidation of LIFO inventory must be disclosed.

Analysis of Inventory

- If finished goods inventory is increasing while raw material and work in progress inventory is decreasing, then this may indicate decreasing demand for the product.
- If raw material and work in progress inventory is increasing, then this may indicate increasing demand for the product.
- If finished goods inventory is increasing at a greater rate than increases in sales, then this may indicate decreasing demand for the product.